

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Geology Group A Higher and Technical Education Department
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4410092094373 Question Type : MCQ

Which dominant dissolved inorganic carbon species primarily buffers the pH of seawater?

Options :

1. Calcium ion (Ca^{2+})
2. Sulfate ion (SO_4^{2-})
3. Dissolved carbon dioxide (CO_2)
4. Bicarbonate ion (HCO_3^-)

Question Number : 2 Question Id : 4410092094375 Question Type : MCQ

How does the remineralization of organic matter in deep ocean sediments typically affect the pH of adjacent bottom waters?

Options :

1. It releases CO_2 , decreasing the local pH of bottom waters
2. It neutralizes acidic inputs, stabilizing pH levels
3. It consumes bicarbonate, increasing the pH of pore waters
4. It enhances alkalinity, raising the pH of seawater

Question Number : 3 Question Id : 4410092094333 Question Type : MCQ

Which combination of atmospheric properties primarily determines the density of an air mass?

Options :

1. Air pressure, temperature, and moisture content
2. Wind speed, direction, and cloud cover
3. Moisture content, precipitation, and visibility
4. Altitude, wind direction, and solar radiation

Question Number : 4 Question Id : 4410092094340 Question Type : MCQ

Why does atmospheric pressure typically decrease more rapidly with height in the lower atmosphere than at higher altitudes?

Options :

1. Stronger frictional forces at lower levels
2. Greater density of air molecules near the surface
3. Constant gravitational acceleration
4. Higher ozone concentration at altitude

Question Number : 5 Question Id : 4410092094337 Question Type : MCQ

What is the primary reason for the steep pressure gradient near the eyewall of a tropical cyclone?

Options :

1. Uniform moisture distribution within the cyclone core
2. Slow warming of the surrounding atmospheric air
3. Very low central pressure created by warm-core air
4. Greater surface friction acting on inflowing winds

Question Number : 6 Question Id : 4410092094415 Question Type : MCQ

The Aravalli Supergroup is predominantly exposed in which Indian state?

Options :

1. Rajasthan
2. Madhya Pradesh
3. Chhattisgarh
4. Kerala

Question Number : 7 Question Id : 4410092094416 Question Type : MCQ

Based on its overall mineralogical composition, what feature justifies classifying peridotite as an ultramafic rock?

Options :

1. High proportion of quartz
2. High content of amphibole minerals
3. High proportion of feldspar minerals
4. Very high content of olivine and pyroxene minerals

Question Number : 8 Question Id : 4410092093854 Question Type : MCQ

Which observation in fluid inclusions would most strongly suggest mixing between magmatic and meteoric fluids during ore deposition?

Options :

1. Consistent inclusion size and similar phase proportions
2. Variable salinity and heterogeneous volatile contents
3. Identical crystal morphology and mineral association
4. Uniform homogenization temperature and fluid density

Question Number : 9 Question Id : 4410092093858 Question Type : MCQ

In magmatic ore formation, which process best explains the early concentration of chromite in layered mafic intrusions?

Options :

1. Metasomatic replacement of silicate minerals
2. Gravitational compaction after solidification
3. Crystal settling during magma differentiation
4. Hydrothermal alteration of country rocks

Question Number : 10 Question Id : 4410092093853 Question Type : MCQ

A mining operation encounters titaniferous magnetite ore that has undergone high-grade metamorphism. Which effect is most likely to alter the ore's extractive metallurgy characteristics?

Options :

1. Transformation of magnetite into hematite matrix
2. Complete replacement of magnetite by garnet
3. Exsolution of ilmenite lamellae within magnetite grains
4. Uniform grain size reduction of all ore minerals

Question Number : 11 Question Id : 4410092093860 Question Type : MCQ

In a workplace scenario, you are presented with an ore sample exhibiting unexpected low reflectance and vivid internal reflections. The mine engineer suspects a processing error. Which mineral inclusion is most likely causing these optical features in the sample?

Options :

1. Sphalerite
2. Chalcopyrite
3. Pyrite
4. Galena

Question Number : 12 Question Id : 4410092093862 Question Type : MCQ

In evaluating the environmental impact of nuclear energy resource development, which geochemical process is most critical in determining the mobility of radionuclides in subsurface environments?

Options :

1. Redox reactions are most critical in controlling radionuclide mobility.
2. Volatilization is most critical in controlling radionuclide mobility.
3. Mechanical dispersion is most critical in controlling radionuclide mobility.
4. Thermal gradient is most critical in controlling radionuclide mobility.

Question Number : 13 Question Id : 4410092094380 Question Type : MCQ

When interpreting electromagnetic survey data, what does a significant phase shift between the primary and secondary fields most commonly indicate?

Options :

1. Detection of a resistive mineral layer
2. Presence of a conductive subsurface body
3. Influence of shallow groundwater table
4. Effect of surface topography

Question Number : 14 Question Id : 4410092094377 Question Type : MCQ

Which physical property is primarily measured in the Time-Domain Induced Polarization (TDIP) method?

Options :

1. Electrical resistivity of geological formations
2. Dielectric constant of sediments
3. Magnetic susceptibility of rocks
4. Chargeability of subsurface materials

Question Number : 15 Question Id : 4410092094201 Question Type : MCQ

Which physical characteristic of aggregates directly influences their resistance to crushing under load?

Options :

1. Shape
2. Color
3. Hardness
4. Strength

Question Number : 16 Question Id : 4410092094200 Question Type : MCQ

How does the orientation of bedding planes relative to a slope affect the likelihood of mass movement?

Options :

1. Bedding planes perpendicular to the slope reduce risk
2. Bedding planes dipping into the slope increase instability
3. Flat-lying bedding planes at the slope base encourage movement
4. Bedding planes dipping with the slope promote instability

Question Number : 17 Question Id : 4410092094195 Question Type : MCQ

Which method yields the most comprehensive understanding of subsurface geology for dam foundation assessment?

Options :

1. Applying one geophysical technique in isolation
2. Integrating borehole data with geophysical survey results
3. Interpreting aerial photographs of surface features
4. Conducting surface geological mapping alone

Question Number : 18 Question Id : 4410092094199 Question Type : MCQ

A hydroelectric dam is planned in a valley with a history of deep-seated landslides. Which combination of geological features would most seriously complicate the long-term stability of the dam site?

Options :

1. Well-cemented conglomerate layers
2. Thin and discontinuous limestone interbeds
3. Isolated boulders in surface colluvium deposits
4. Thick sequences of weak clay layers beneath the valley floor

Question Number : 19 Question Id : 4410092094344 Question Type : MCQ

In the context of water and energy resources, what is a key effect of energy degradation on hydropower systems?

Options :

1. Improved resource renewal rates
2. Higher energy output from turbines
3. Increased stored water in reservoirs
4. Lower efficiency in energy conversion

Question Number : 20 Question Id : 4410092094342 Question Type : MCQ

Which integrated strategy would best address both resource depletion and local environmental degradation in mining areas?

Options :

1. Providing tax incentives while expanding mineral extraction activities
2. Relocating nearby communities without improving mining operations
3. Enforcing sustainable mining with community rehabilitation initiatives
4. Focusing solely on identifying new mineral reserve locations

Question Number : 21 Question Id : 4410092094348 Question Type : MCQ

Which primary factor contributes to the intensification of tropical cyclones before they make landfall along coastal regions?

Options :

1. High sea surface temperatures
2. Increased land vegetation
3. Low atmospheric humidity
4. Strong upper-level winds

Question Number : 22 Question Id : 4410092094351 Question Type : MCQ

Which combination of factors is most likely to trigger a rapid, large-scale landslide in a tectonically active, deforested area during the rainy season?

Options :

1. Gentle slopes, low rainfall, thick vegetation, no seismic disturbance
2. Moderate slopes, sporadic rainfall, sandy soils, stable tectonics
3. Steep slopes, intense rainfall, shallow soil cover, recent seismic activity
4. Flat terrain, moderate rainfall, deep soil cover, absence of earthquakes

Question Number : 23 Question Id : 4410092094386 Question Type : MCQ

Which analytical technique is most commonly used to determine trace element concentrations in soil samples during geochemical exploration?

Options :

1. Atomic Absorption Spectroscopy
2. Neutron Activation Analysis
3. Inductively Coupled Plasma Mass Spectrometry
4. X-Ray Fluorescence

Question Number : 24 Question Id : 4410092094385 Question Type : MCQ

Which principle underlies the use of indicator plant species in geobotanical mineral exploration?

Options :

1. Plants always avoid mineralized soils
2. Certain plants accumulate specific elements from underlying ore bodies
3. Most plants change color near mineralization zones
4. Random plant growth patterns indicate mineralization

Question Number : 25 Question Id : 4410092093876 Question Type : MCQ

When assessing the migration of trace elements in a basaltic melt, which property is crucial for predicting their behavior during cooling and solidification?

Options :

1. Elemental diffusivity
2. Crystal habit
3. Oxygen fugacity
4. Partition coefficient

Question Number : 26 Question Id : 4410092093877 Question Type : MCQ

Which thermodynamic parameter is adjusted to account for non-ideal solution behavior in geochemical reactions involving aqueous ions?

Options :

1. Enthalpy is recalculated to include excess mixing energy.
2. Activity coefficient is used to modify the concentration.
3. Entropy is adjusted to reflect disorder changes.
4. Partial pressure is substituted for concentration.

Question Number : 27 Question Id : 4410092093872 Question Type : MCQ

In a surface water monitoring program, a geochemist observes diurnal fluctuations in dissolved oxygen and pH. What process best accounts for this observation based on applied geochemistry?

Options :

1. Thermal stratification of water column
2. Photosynthesis and respiration of aquatic plants
3. Precipitation and dissolution of minerals
4. Groundwater inflow into the river

Question Number : 28 Question Id : 4410092093875 Question Type : MCQ

In evaluating rare earth element substitution in rock-forming minerals, which atomic property most effectively explains their similar behavior during geochemical processes?

Options :

1. Close electron affinity values
2. Comparable ionic radii and identical charge
3. Similar nuclear mass and neutron count
4. Equal number of protons in the nucleus

Question Number : 29 Question Id : 4410092093878 Question Type : MCQ

In a contact metamorphic aureole, why might nucleation of new silicate minerals predominantly occur along grain boundaries rather than within crystal lattices?

Options :

1. Grain boundaries provide lower energy sites for nucleation
2. The chemical potential is uniform across the rock
3. Crystal lattice defects are more abundant within grains
4. Localized temperature peaks occur at grain boundaries

Question Number : 30 Question Id : 4410092094394 Question Type : MCQ

Which region in India generally experiences the highest annual temperature range due to its continental location?

Options :

1. Northwestern plains of India
2. Peninsular plateau region
3. Eastern coastal belt
4. Northeastern hilly states

Question Number : 31 Question Id : 4410092094390 Question Type : MCQ

Which policy approach would most effectively address both desertification in Rajasthan and groundwater depletion in Tamil Nadu simultaneously?

Options :

1. Enforcing stricter pollution controls in industries
2. Expanding afforestation programs in all states
3. Implementing integrated watershed management across regions
4. Constructing large dams for irrigation improvement

Question Number : 32 Question Id : 4410092094215 Question Type : MCQ

Which landform is produced by glacial deposition and has an elongated shape along the direction of ice movement?

Options :

1. Esker
2. Moraine
3. Cirque
4. Drumlin

Question Number : 33 Question Id : 4410092094208 Question Type : MCQ

When examining sandstone landscapes, which landform is typically produced by differential erosion due to the contrasting hardness of rock layers?

Options :

1. Cuesta ridges
2. Inselberg hills
3. U-shaped valleys
4. Floodplain terraces

Question Number : 34 Question Id : 4410092094205 Question Type : MCQ

A delta undergoing rapid progradation exhibits which dominant depositional process at its seaward margin?

Options :

1. Channel abandonment with mudflat development
2. Distributary avulsion leading to inland sedimentation
3. Wave reworking causing lateral retreat of the delta front
4. Seaward growth through mouth-bar deposition

Question Number : 35 Question Id : 4410092094216 Question Type : MCQ

Which combination of oceanic features would provide the strongest evidence for a past episode of plate divergence?

Options :

1. An oceanic trench with adjacent accretionary wedge
2. A mid-ocean ridge with symmetrical magnetic anomalies
3. A submarine canyon crossing the continental shelf
4. A group of seamounts with flat tops

Question Number : 36 Question Id : 4410092094399 Question Type : MCQ

Which parameter primarily controls the **amplitude** of the gravity anomaly produced by a buried sphere, assuming its size and depth are known?

Options :

1. Depth to the center of the sphere
2. Strike direction of the anomaly
3. Inclination of the anomaly
4. Density contrast between sphere and host rock

Question Number : 37 Question Id : 4410092094396 Question Type : MCQ

During underwater magnetic data acquisition, what is a primary benefit of using a towed magnetometer array instead of a single sensor unit?

Options :

1. Minimizes ship-induced magnetic noise
2. Collects higher-resolution vertical field profiles
3. Enables spatial coverage for detecting localized anomalies
4. Reduces overall equipment weight in deployment

Question Number : 38 Question Id : 4410092094402 Question Type : MCQ

Why is the **Bouguer correction** applied after the **free-air correction** in gravity data processing?

Options :

1. To account for changes in Earth's curvature at different latitudes
2. To remove the gravitational effect of surface mass between the station and sea level
3. To compensate for magnetic field variations at the survey site
4. To ensure latitude corrections are not affected by elevation

Question Number : 39 Question Id : 4410092094403 Question Type : MCQ

Which factor most directly causes the geoid to deviate from the reference ellipsoid in Earth's gravity field studies?

Options :

1. Lateral variations in density within the Earth's crust and mantle
2. Surface topography differences at sea level
3. Constant rotation speed of the Earth
4. Uniform thickness of the Earth's crust

Question Number : 40 Question Id : 4410092094023 Question Type : MCQ

A hydrogeologist finds that the measured hydraulic conductivity in a fractured granite aquifer is much higher than predicted by Darcy's law applied to the rock matrix. What is the most likely explanation?

Options :

1. Temperature fluctuations have altered the groundwater properties
2. Darcy's law is not strictly valid in fractured systems due to preferential flow paths
3. The aquifer matrix is fully saturated during testing
4. Measurement instruments underestimated the pressure gradient

Question Number : 41 Question Id : 4410092094020 Question Type : MCQ

When applying the unit hydrograph method to a large heterogeneous basin, what is a significant limitation of this approach?

Options :

1. Requirement for continuous hourly rainfall data to construct hydrographs
2. Inability to separate baseflow from total streamflow response
3. Applicability is limited to arid regions with simple land cover
4. Spatial variability of rainfall and catchment characteristics affects accuracy

Question Number : 42 Question Id : 4410092094012 Question Type : MCQ

If a coastal city increases groundwater extraction and causes the freshwater-saltwater interface to move inland, which management action would best restore the interface to its original position?

Options :

1. Installing deeper production wells in the same aquifer
2. Lining surface water canals with impermeable materials
3. Reducing groundwater pumping rates below recharge levels
4. Constructing multiple observation wells near the interface

Question Number : 43 Question Id : 4410092094013 Question Type : MCQ

In the context of groundwater provinces, what hydrological characteristic distinguishes the Indo-Gangetic plain from the Deccan Plateau?

Options :

1. Continuous unconfined aquifer systems
2. Dominance of karstic conduit flow
3. Prevalence of saline water intrusion
4. High semi-arid recharge rates

Question Number : 44 Question Id : 4410092094421 Question Type : MCQ

Which stress regime is most likely to produce normal faults in rocks?

Options :

1. Hydrostatic pressure
2. Tensional stress
3. Compressional stress
4. Shear stress

Question Number : 45 Question Id : 4410092094425 Question Type : MCQ

Which property best differentiates an isoclinal fold from other fold types in the context of limb orientation?

Options :

1. Both limbs dip at low angles in opposite direction
2. Both limbs are parallel or nearly parallel in orientation
3. Both limbs show steep dips around a vertical fold axis
4. Both limbs dip away from the hinge in opposite directions

Question Number : 46 Question Id : 4410092093892 Question Type : MCQ

Which ocean floor tectonic domain serves as the primary location for the recycling of oceanic lithosphere back into the mantle?

Options :

1. Abyssal plain
2. Seamount
3. Oceanic trench
4. Mid-ocean ridge

Question Number : 47 Question Id : 4410092093887 Question Type : MCQ

Which combination of temperature and salinity conditions is most favorable for the generation of North Atlantic Deep Water (NADW)?

Options :

1. Low temperature and low salinity in surface waters
2. High temperature and high salinity in surface waters
3. High temperature and low salinity in surface waters
4. Low temperature and high salinity in surface waters

Question Number : 48 Question Id : 4410092093889 Question Type : MCQ

An oceanographic expedition detects a subsurface salinity maximum at intermediate depths in the subtropical North Atlantic. Which water mass is primarily responsible for this feature?

Options :

1. Antarctic Intermediate Water
2. Mediterranean Overflow Water
3. North Atlantic Deep Water
4. Labrador Sea Water

Question Number : 49 Question Id : 4410092093894 Question Type : MCQ

A paleoceanographer is modeling Cenozoic sea level in response to both Antarctic glaciation and variations in seafloor spreading rate. Which combination of outcomes is most likely when both ice volume increases and ridge volume decreases during the same interval?

Options :

1. No significant change occurs as the processes counterbalance each other
2. Global sea level falls due to increased ice storage and reduced ridge elevation
3. Sea level rises due to enhanced seafloor spreading despite ice accumulation
4. Sea level rises because ridge subsidence offsets glacial ice growth

Question Number : 50 Question Id : 4410092093895 Question Type : MCQ

Suppose the modern Drake Passage was suddenly closed. Which scenario would best describe the potential impact on Southern Ocean circulation and Antarctic climate?

Options :

1. Intensified Antarctic glaciation with enhanced polar cooling
2. Unchanged Southern Ocean circulation and Antarctic climate
3. Strengthened Antarctic Circumpolar Current and polar cooling
4. Weakened Antarctic Circumpolar Current and polar warming

Question Number : 51 Question Id : 4410092093902 Question Type : MCQ

In the study of magma mixing, which textural feature is most indicative of incomplete chemical homogenization between two magmas?

Options :

1. Uniform crystal size throughout the rock
2. Completely homogenous groundmass texture
3. Absence of zoned minerals in phenocrysts
4. Presence of mafic enclaves within a felsic host

Question Number : 52 Question Id : 4410092093900 Question Type : MCQ

In the context of igneous classification history, what was a major limitation of early chemical classification schemes before the development of the IUGS system?

Options :

1. They often excluded mineralogical and textural information from classification
2. They often emphasized geographic location in rock classification schemes
3. They often relied on subjective color features for classification
4. They often required metamorphic evidence for rock classification

Question Number : 53 Question Id : 4410092093909 Question Type : MCQ

Which structural characteristic is shared by both amphibole and pyroxene groups but differs in their arrangement?

Options :

1. Three-dimensional frameworks of interconnected tetrahedra
2. Isolated tetrahedral units with no shared atoms
3. Chains of tetrahedra linked by shared oxygen atoms
4. Closed rings of tetrahedra sharing two oxygens

Question Number : 54 Question Id : 4410092093901 Question Type : MCQ

Which textural evidence in an igneous rock most convincingly points to prior silicate melt immiscibility during crystallization?

Options :

1. Globular inclusions of one silicate melt within another silicate matrix
2. Sieve-textured plagioclase crystals within the igneous rock matrix
3. Exsolution lamellae developed within olivine or pyroxene crystals
4. Interlocking plagioclase and pyroxene crystals within the rock

Question Number : 55 Question Id : 4410092093904 Question Type : MCQ

A regionally metamorphosed terrain shows a sequence of rocks progressing from lawsonite-glaucophane assemblages to garnet-omphacite assemblages. What tectonic and metamorphic evolution does this sequence most likely reflect?

Options :

1. Collision-related contact metamorphism
2. Stable continental crust evolution
3. Exhumation of high-pressure rocks
4. Subduction of oceanic crust to greater depths

Question Number : 56 Question Id : 4410092094357 Question Type : MCQ

Which method is most commonly used as the primary mechanical technique to contain and remove oil from marine oil spills?

Options :

1. Incineration of contaminated water
2. Application of chemical dispersants to break up oil
3. Use of booms and skimmers to contain and remove oil
4. Dilution of oil with excess seawater

Question Number : 57 Question Id : 4410092094358 Question Type : MCQ

What distinguishes North Atlantic Deep Water from Antarctic Intermediate Water in terms of origin and properties?

Options :

1. North Atlantic Deep Water forms at higher latitudes and is colder and saltier
2. Antarctic Intermediate Water forms at higher latitudes and is colder and saltier
3. North Atlantic Deep Water forms at lower latitudes and is fresher and warmer
4. Antarctic Intermediate Water forms at lower latitudes and is saltier and colder

Question Number : 58 Question Id : 4410092093926 Question Type : MCQ

A paleontologist examining microfossils in sedimentary rock identifies tiny, single-celled organisms with ornate silica skeletons. Which microfossil type is being described?

Options :

1. Diatoms
2. Radiolarians
3. Coccolithophores
4. Foraminifera

Question Number : 59 Question Id : 4410092093925 Question Type : MCQ

In paleoecological studies, which method is most effective for determining the dominant vegetation type in a sedimentary sequence containing plant fossils?

Options :

1. Measuring stem diameters in petrified wood
2. Analysis of pollen and spores in sediment samples
3. Counting leaf impressions in rock layers
4. Examining charcoal fragments in deposits

Question Number : 60 Question Id : 4410092093923 Question Type : MCQ

In a paleoclimatic study, the relative abundance of warm-water ostracod species declines sharply in a lake core, while cold-adapted ostracod species increase. Which paleoclimatic scenario does this microfossil shift best support?

Options :

1. Rapid eutrophication from increased nutrient input.
2. A shift toward more saline and evaporative lake phases.
3. A transition to cooler and possibly deeper lake conditions.
4. A period of abrupt lake desiccation and exposure.

Question Number : 61 Question Id : 4410092093924 Question Type : MCQ

In reconstructing the trophic structure of a Jurassic marine community, a paleoecologist integrates evidence from coprolites, tooth wear patterns, and gastrolith presence. Which interpretation best utilizes these data sets together?

Options :

1. Mapping the geographic distribution of fossil localities
2. Determining sedimentation rates from fossil occurrence
3. Estimating the relative ages of individual fossil species
4. Inferring predator-prey and feeding strategies

Question Number : 62 Question Id : 4410092093927 Question Type : MCQ

A deep oceanic sediment core records microfossil assemblages that indicate repeated alternations between warm and cold water species, aligned with evidence of tectonic rifting. How can this pattern be interpreted in terms of sea floor tectonism?

Options :

1. Short-term climate events produced brief faunal changes
2. Volcanic eruptions led to single faunal turnovers
3. Stable sedimentation maintained consistent assemblages
4. Tectonic rifting altered oceanic circulation, causing periodic faunal shifts

Question Number : 63 Question Id : 4410092094361 Question Type : MCQ

What is the primary objective of Integrated Coastal Zone Management (ICZM)?

Options :

1. Ensuring sustainable use of coastal resources
2. Focusing on the interests of a single stakeholder group
3. Prioritizing tourism development over other coastal activities
4. Promoting economic expansion without environmental safeguards

Question Number : 64 Question Id : 4410092094360 Question Type : MCQ

Which physical characteristic is primarily used to classify an estuary as a salt-wedge type?

Options :

1. Prevalence of wave-driven sediment deposition
2. Dominance of river discharge over tidal action
3. Strong wind-induced vertical mixing
4. Tidal mixing exceeding river flow

Question Number : 65 Question Id : 4410092094367 Question Type : MCQ

Which mechanism best explains the maintenance of Antarctic Bottom Water's distinct characteristics despite widespread ocean mixing?

Options :

1. Equatorial upwelling events
2. Wind-driven surface mixing
3. Persistent high-latitude brine rejection
4. Hydrothermal vent activity

Question Number : 66 Question Id : 4410092227651 Question Type : MCQ

Which feature provides primary support for the theory of sea floor spreading?

Options :

1. Varying ocean water salinity levels
2. Irregular trench distribution in oceans
3. Symmetrical magnetic patterns on ocean floors
4. Irregular sediment layers across ridges

Question Number : 67 Question Id : 4410092094407 Question Type : MCQ

How does low thermal conductivity of crustal rocks affect the geothermal gradient?

Options :

1. It decreases the geothermal gradient.
2. It increases the geothermal gradient.
3. It has no effect on the geothermal gradient.
4. It causes geothermal gradients to vary randomly.

Question Number : 68 Question Id : 4410092094031 Question Type : MCQ

Which structural feature is most commonly associated with Precambrian mobile belts?

Options :

1. Massive sedimentary basins with little deformation
2. Extensive horizontal lava flows and basalt layers
3. Complexly folded and faulted metamorphic rock sequences
4. Large plutonic batholiths intruding sedimentary rocks

Question Number : 69 Question Id : 4410092094025 Question Type : MCQ

Which environmental factor is most likely to have triggered repeated expansion and contraction of the Precambrian cryosphere?

Options :

1. Increased meteorite flux and crustal impacts
2. Episodes of oceanic anoxia and biological turnover
3. Plate tectonic rearrangements and mountain building
4. Variations in solar luminosity and greenhouse gas concentrations

Question Number : 70 Question Id : 4410092094032 Question Type : MCQ

Considering the crustal evolution of the Indian shield, which Proterozoic basin provides the most complete record of basin subsidence, sedimentation, and tectonic reactivation from the Paleoproterozoic to Neoproterozoic?

Options :

1. Kaladgi Basin
2. Chhattisgarh Basin
3. Vindhyan Supergroup Basin
4. Cuddapah Basin

Question Number : 71 Question Id : 4410092094041 Question Type : MCQ

During radiocarbon dating, why must the initial ratio of Carbon-14 to Carbon-12 in the atmosphere be considered?

Options :

1. It ensures fossil preservation for dating
2. It provides direct evidence for the Quaternary Period
3. It is needed to accurately calculate the elapsed time since death
4. It prevents contamination of the sample

Question Number : 72 Question Id : 4410092094033 Question Type : MCQ

What does the prevalence of stabilized dune fields in the Indian Thar Desert during the Early Holocene suggest about past climatic conditions?

Options :

1. Reduced sediment supply from adjacent mountain sources
2. Lower wind strength associated with regional tectonic uplift
3. Increased human settlement leading to land stabilization
4. Increased vegetation cover due to higher precipitation regimes

Question Number : 73 Question Id : 4410092094036 Question Type : MCQ

In the context of Indian Quaternary fluvial records, which approach is most robust for reconstructing paleoclimate at centennial-scale resolution?

Options :

1. Stable isotope analysis of calcrete nodules within alluvial profiles
2. Radiocarbon dating of terraces and analysis of terrace heights
3. Heavy mineral analysis of channel sands from multiple terrace levels
4. Multi-proxy analysis integrating sedimentology, geochemistry, and fossil pollen data

Question Number : 74 Question Id : 4410092094040 Question Type : MCQ

When interpreting Quaternary cryospheric records, what methodological step is most critical to minimize post-depositional alteration of stable isotope signals?

Options :

1. Increasing sample volume to average out noise
2. Focusing only on the deepest sections of ice cores
3. Analyzing only dust-rich segments for isotope data
4. Selection of core sections away from melt layers

Question Number : 75 Question Id : 4410092094051 Question Type : MCQ

Which advancement in GPS technology allows for improved geological interpretation by reducing atmospheric signal errors during remote sensing surveys?

Options :

1. Deployment of single-frequency receivers
2. Exclusive reliance on post-processing software
3. Increasing the number of satellites in view
4. Utilization of dual-frequency measurements

Question Number : 76 Question Id : 4410092094049 Question Type : MCQ

Which method best supports quantitative geological mapping from rasterized aerial photographs in GIS?

Options :

1. Merging vector data with raster images
2. Adding ground control points to the raster image
3. Tracing lithological boundaries manually on screen
4. Classifying pixel values based on spectral characteristics

Question Number : 77 Question Id : 4410092094045 Question Type : MCQ

When performing high-precision DGPS surveys, which factor most critically affects the transformation of coordinates from a local datum to WGS84?

Options :

1. The type of GPS receiver model used in the survey
2. The grid size used in the coordinate system definition
3. The number of visible satellites during DGPS operation
4. The accuracy of the datum transformation between the local datum and WGS84

Question Number : 78 Question Id : 4410092094056 Question Type : MCQ

When interpreting multispectral satellite imagery for lithological boundaries in a region with extensive weathering, which approach is most likely to improve the accuracy of geological mapping?

Options :

1. Integrating multispectral data with field validation and ancillary geological/topographic data
2. Applying high-pass filtering to enhance boundaries
3. Using single-band infrared images for lithological discrimination
4. Exclusive use of radar backscatter for lithological identification

Question Number : 79 Question Id : 4410092094224 Question Type : MCQ

Which instrument is part of the GPM Core Observatory for measuring precipitation?

Options :

1. Microwave radiometer
2. Ultraviolet sensor
3. Dual-frequency radar
4. Infrared spectrometer

Question Number : 80 Question Id : 4410092094217 Question Type : MCQ

When designing meteorological observation strategies, why is the use of geostationary satellites preferred for rapid storm tracking over oceanic regions?

Options :

1. They provide detailed subsurface ocean measurements
2. They collect data at multiple orbital inclinations
3. They perform frequent low altitude passes for oceanic analysis
4. They maintain constant surveillance over large oceanic areas

Question Number : 81 Question Id : 4410092094223 Question Type : MCQ

When integrating multiple satellite products to analyze climate trends, what strategy maximizes the validity of detected changes?

Options :

1. Applying temporal smoothing techniques
2. Focusing on a single climate variable
3. Cross-verification using independent datasets
4. Using data from a specific region

Question Number : 82 Question Id : 4410092094228 Question Type : MCQ

How do geostationary satellite observations enhance real-time wind analysis compared to polar-orbiting satellites in operational meteorology?

Options :

1. Higher spatial resolution for wind retrieval
2. Advanced spectral capabilities for wind assessment
3. Global coverage with infrequent revisits
4. Continuous monitoring of atmospheric motion

Question Number : 83 Question Id : 4410092094071 Question Type : MCQ

Which relatively recent stratigraphic approach uses distinctive elemental or isotopic signatures to correlate sedimentary sequences across continents?

Options :

1. Chemostratigraphy
2. Biostratigraphy
3. Lithostratigraphy
4. Magnetostratigraphy

Question Number : 84 Question Id : 4410092094058 Question Type : MCQ

What is the most likely diagenetic alteration observed in carbonate rocks subjected to meteoric phreatic conditions?

Options :

1. Precipitation of glauconite in pore spaces
2. Development of secondary porosity through dissolution
3. Dolomitization by magnesium-rich fluids
4. Replacement by anhydrite minerals

Question Number : 85 Question Id : 4410092094063 Question Type : MCQ

Which principle limits the use of benthic foraminifera for global biostratigraphic correlation compared to planktonic foraminifera?

Options :

1. Benthic foraminifera have limited geographic distribution due to environmental preferences
2. Benthic foraminifera are absent from deep-sea sediment cores
3. Benthic foraminifera evolve too rapidly for detailed zonation
4. Benthic foraminifera rarely fossilize in marine sediments

Question Number : 86 Question Id : 4410092094066 Question Type : MCQ

Which field characteristic would most readily help a geologist identify a lithostratigraphic boundary between two sedimentary units?

Options :

1. Sharp change in lithology, grain size and sedimentary structures at a distinct contact
2. Subtle mineralogical change detectable only under microscope
3. Gradual transition in rock color with little change in texture
4. Presence of a unique fossil assemblage across several beds

Question Number : 87 Question Id : 4410092094069 Question Type : MCQ

In the Cambay Basin, the Paleocene–Early Eocene succession is characterized by relatively high sediment accumulation and subsidence rates. Which primary geotectonic process best explains this pattern?

Options :

1. Widespread volcanic activity
2. Eustatic sea-level fluctuations
3. Active fault-controlled basin subsidence
4. Post-rift thermal subsidence

Question Number : 88 Question Id : 4410092094428 Question Type : MCQ

When the seismic velocities of the layers are known, the crossover distance on a seismic refraction travel-time graph is primarily used to estimate which parameter?

Options :

1. The seismic source location
2. The frequency of the seismic source
3. The density contrast between layers
4. The depth to the refractor

Question Number : 89 Question Id : 4410092094427 Question Type : MCQ

During interpretation, why is the velocity model critical when mapping horizontal interfaces in seismic sections?

Options :

1. It sets the recording window duration
2. It determines seismic amplitude scaling
3. It ensures accurate depth conversion of reflection times
4. It enhances continuity of reflection events

Question Number : 90 Question Id : 4410092094434 Question Type : MCQ

Which feature enables broad band seismometers to record both low and high frequency seismic waves effectively?

Options :

1. Wide frequency response
2. Narrow frequency response
3. High sensitivity
4. Simple construction

Question Number : 91 Question Id : 4410092094433 Question Type : MCQ

How does the seismic gap hypothesis assist in earthquake hazard assessment?

Options :

1. By predicting the exact timing of future earthquakes along fault segments
2. By analyzing only the distribution of aftershocks along fault segments
3. By examining only the occurrence of deep earthquakes along fault segments
4. By identifying fault segments with accumulated stress from long earthquake gaps

Question Number : 92 Question Id : 4410092094080 Question Type : MCQ

How can fluids influence strain localization in deforming rocks?

Options :

1. Fluids enhance strain localization by promoting mineral reactions.
2. Fluids block strain localization and maintain homogeneity.
3. Fluids only affect brittle strain localization, not ductile zones.
4. Fluids distribute strain evenly throughout the rock mass.

Question Number : 93 Question Id : 4410092094075 Question Type : MCQ

Which flow pattern is dominant in the formation of S-C fabrics within mylonites?

Options :

1. Pure shear flow
2. Simple shear flow
3. Cataclastic flow
4. Coaxial shortening flow

Question Number : 94 Question Id : 4410092094077 Question Type : MCQ

A geologist observes recumbent folds and flow structures within a slope failure deposit. What is the most plausible interpretation of their origin?

Options :

1. Magmatic intrusion causing local folding
2. Syn-depositional volcanic activity and uplift
3. Gravity-induced deformation during mass wasting
4. Deep-seated tectonic compression and thrusting

Question Number : 95 Question Id : 4410092094072 Question Type : MCQ

A field study reveals overturned folds, penetrative cleavage, and mylonitic shear zones in a mountain belt. What tectonic environment most likely produced this association?

Options :

1. Thin-skinned thrusting in a foreland fold belt
2. Large-scale crustal extension in a rift setting
3. Strike-slip faulting along a transform boundary
4. Thick-skinned crustal shortening in a collisional orogen

Question Number : 96 Question Id : 4410092094081 Question Type : MCQ

Which structural feature is most commonly associated with a dextral strike-slip terrain in a brittle upper crustal environment?

Options :

1. Vertical tension fractures near the fault
2. Antithetic faults parallel to the main fault
3. Riedel shears trending obliquely to the main fault
4. Symmetric folds with no systematic relationship to the fault

Question Number : 97 Question Id : 4410092094440 Question Type : MCQ

What is the Widmanstätten pattern observed in meteorites?

Options :

1. A thin glassy layer formed on atmospheric entry
2. A unique interlocking structure of nickel-iron crystals
3. A weathered surface formed by terrestrial oxidation
4. A circular depression caused by surface impact

Question Number : 98 Question Id : 4410092094441 Question Type : MCQ

How does the chemical composition of the Earth's mantle differ from that of the crust?

Options :

1. The mantle contains more potassium and less iron than the crust.
2. The mantle contains more sodium and less magnesium than the crust.
3. The mantle contains more magnesium and less silicon than the crust.
4. The mantle contains more aluminum and less iron than the crust.

Question Number : 99 Question Id : 4410092094450 Question Type : MCQ

In a consolidated sandstone, what does a decrease in sonic log travel time indicate about the formation porosity?

Options :

1. Increase in effective porosity
2. Decrease in porosity
3. Increase in porosity
4. No significant change in porosity

Question Number : 100 Question Id : 4410092094448 Question Type : MCQ

Which physical property is directly measured by a gamma ray tool during logging while drilling?

Options :

1. Formation density
2. Formation resistivity
3. Natural radioactivity
4. Formation porosity